

Input: IOCG mineral-system criterion →

Output: source passages + graph records

a Relevance by geological criterion

Directly relevant passages (0–5)

		BM25	MiniLM	MPNet	KG	BM25 + KG	4-way fusion
S1	Mafic–ultramafic rock sources	3	3	3	3	4	5
S2	Metasomatized mantle sources	3	3	1	3	3	3
S3	Calc-silicate source proxies	0	1	3	1	0	3
S3b	Evaporite-bearing rocks	4	5	4	4	3	4
D1	High-temperature felsic melts	2	2	1	4	2	3
D2	Mafic–ultramafic magmatism	1	3	2	2	1	3
P1	Crustal-scale weak zones	4	3	5	3	4	5
P2	Lithospheric block boundaries	4	2	1	4	4	3
P3	Crustal conductivity gradients	4	3	4	2	3	4
P4	Craton edges (lithosphere depth)	1	2	2	1	1	1
P5	Crustal thickness	1	2	0	1	2	3
P6	Deformed extensional basins	4	2	2	2	3	4
P7	Sutures (Sm–Nd gradients)	3	0	0	1	3	0
T1	Fe-oxide depositional gradients	1	1	2	0	1	1

b Mean retrieval quality

	Graded P@5	nDCG@5
BM25	0.679	0.694
MiniLM	0.650	0.664
MPNet	0.557	0.578
KG matches	0.586	0.575
BM25 + KG	0.679	0.691
4-way fusion	0.757	0.767

c Complete coverage (%)

2 criteria	3 criteria	4 criteria
 84	 78	 75
 89	 87	 86
 78	 74	 71
 57	 34	 0
 84	 78	 75
 70	 45	 0

Top 3 passages per criterion

S Sources D Drivers P Pathways / architecture T Deposition